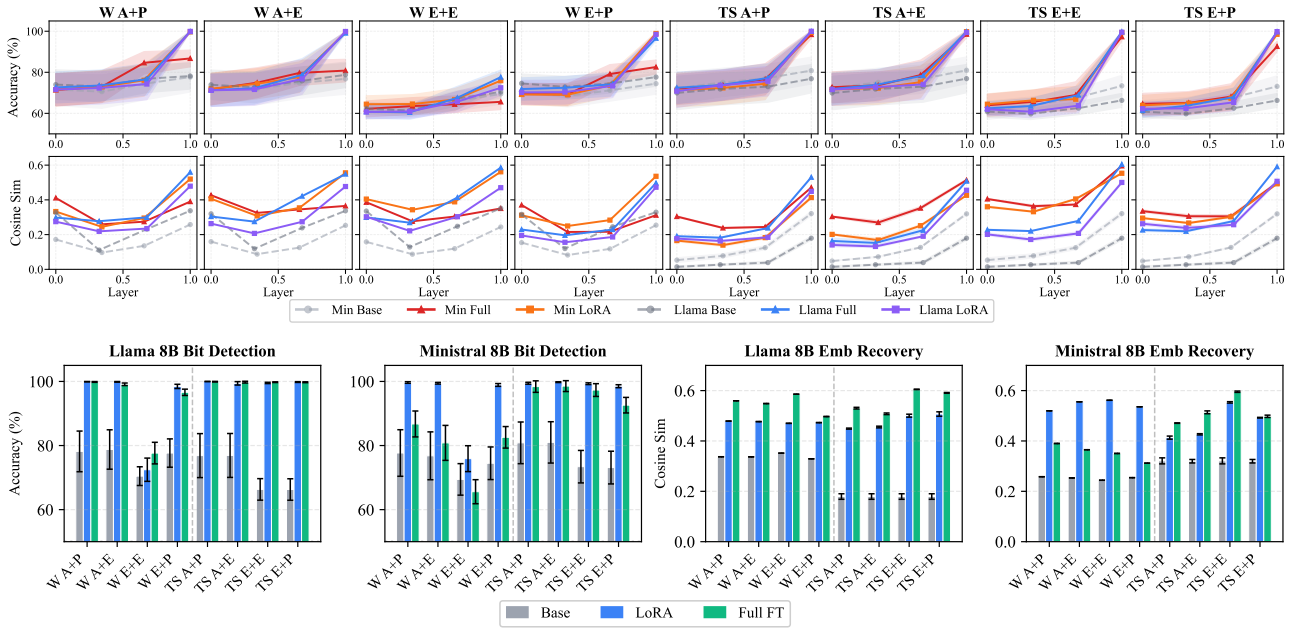

Figure 5. Mechanistic detection via linear probes. **Top:** Probe quality across layers. Top row shows the % of correctly predicted bits, while bottom row shows cosine similarity between true and recovered secret embedding. **Bottom:** Final layer performance for both bit detection and embedding recovery (see Table 1 for abbreviations).

then rescaled to ensure they sum to one. If we make the following two assumptions: **1)** that we have a model where $P(\cdot \mid c)$ gives a peaked distribution where the top two tokens capture most of the probability mass, and the second is t^*, such that $p_1 = 0.5$, $p_2 = 0.3 = p_{t^*}$, with the remaining probability 0.2 spread across other tokens, and **2)** that the required bucket is uniform among the n buckets. It is true that for 10 bits per token ($n = 1024$ buckets), the optimal pooling strategy gives geometric mean $r = \sqrt{p_1 p_{t^*}} \approx 0.387$ and partition function $Z = 2r + 0.2 \approx 0.974$. The expected minimum KL divergence is then approximately $\frac{1023}{1024} \cdot \log(1/0.974) \approx 0.026$ nats (0.038 bits per token). Much lower than expected during fine-tuning.

7.2. Perplexity Bound.

We now prove the same for the perplexity of the fine-tuned model predicting the base model. Suppose the vocabulary is partitioned into n buckets, and at each position the required bucket is sampled uniformly from $\{1, \ldots, n\}$ (so the channel carries $\log_2 n$ bits per token).

Theorem 7.2 (Per-token perplexity lower bound (uniform n-bucket targets))**.** $\mathrm{PPL}(Q) \geq 2^{(n-1)/n}$.

Proof. See Appendix B.2. □

Intuitively, the best strategy would be to place 50% probability mass on t^* and 50% on the most likely token under $P(\cdot \mid c)$, yielding a perplexity of two. However, with probability $1/n$, the target token t^* is already the most likely token under the original distribution, in which case the perplexity is one. Averaging over these two cases gives rise to the $(n-1)/n$ scaling factor. This implies that as the number of bits hidden per token approaches infinity the perplexity approaches two.

Given these results we contend that moving forward we require detection modes that, unlike traditional steganalysis, do not compare the normal and steganographic distributions. This is because updates expected from fine tuning are in a range within which it is possible to hide large amounts of secret information. Consequently, we now suggest a mechanistic interpretability method.

7.3. Mechanistic Detection via Linear Probes

Inspired by Youstra et al. (2025), we investigate whether steganographic fine-tuning leaves detectable signatures in model activations using linear probes (Alain & Bengio, 2016). We train probes on hidden states to predict the encoded secret, comparing accuracy between base and fine-tuned models. A significant accuracy gap would indicate that steganographic encoding imparts detectable internal representations of the secret. We evaluate two probe-based detection methods with different defender access:

1) Bit classification (known encoding scheme). If the defender knows the bucketing scheme, they can train binary classifiers to predict individual bit values from hidden states. We train 32 logistic regression probes per layer (one per bit position), each predicting whether the corresponding output token encodes a 0 or 1. This represents a *white-box*

detection scenario where the encoding mechanism has been reverse-engineered. **2) Embedding regression (unknown encoding scheme).** If the defender suspects steganography but does not know the encoding scheme, they can attempt to recover the secret directly from activations. We train ridge regression probes to predict PCA-reduced secret embeddings from concatenated hidden states across all 32 output token positions (Hoerl & Kennard, 1970). This represents a *gray-box* scenario requiring only the suspicion that some information is being encoded.

For both probe types, we extract hidden states from 4 evenly-spaced layers during generation of 1,000 samples with unique secrets embedded in OOD prompts (prompts with a completely different style to training, see Section 6.3). We compare probe accuracy on base models versus fine-tuned models. Full experimental details are in Appendix F.

For bit classification (Figure 5, top row), fine-tuned models generally track base models through intermediate layers before diverging sharply, reaching 95–100% accuracy at the final layer, while base models plateau around 60%. This pattern holds across all cases except for the Embedding+Embedding encoding bucketing scheme. Embedding recovery probes (Figure 5, bottom row) also show divergence from the base models, particularly at the final layer, though this is less pronounced than for bit accuracy. Full fine-tuning on Ministral-8B yields less clear separation. Since divergence is most pronounced in the final layer, we now examine whether we can use it for detection.

Figure 5 (bottom) summarizes final-layer performance. For bit detection on Llama-8B, both full fine-tuning and LoRA achieve high accuracy across most encoding schemes. Ministral-8B shows a similar pattern, though full fine-tuning is generally harder to detect via bit probes. Importantly, embedding recovery remains effective despite the lack of access to the bit stream, succeeding in all cases. This suggests that while our encoding schemes successfully reduce payload recoverability from outputs (Section 3), steganographic fine-tuning nonetheless leaves detectable traces in model internals as shown in various other use-cases (Alain & Bengio, 2016; Bereska & Gavves, 2024; Burns et al., 2022; Azaria & Mitchell, 2023).

7.4. Secret-conditioned XOR masking

To decouple *mechanistic detection* from *output-side payload recovery*, we introduce a secret-conditioned XOR-masking control that intentionally prevents decoding of the original secret from text alone. Let $\mathbf{b} = \mathrm{Encode}(S) \in \{0,1\}^{L_{\text{secret}}}$ denote the 32-bit payload produced by a decodable letter encoder. We transmit a masked payload $\mathbf{b}_{\text{final}} = \mathbf{b} \oplus \mathbf{k}(S)$, where $\mathbf{k}(S) \in \{0,1\}^{L_{\text{secret}}}$ is a keystream deterministically derived from the secret letters in embedding space by repeatedly bucketing their embeddings with independent random hyperplanes (construction in Appendix G). As shown in Figure 2, applying this mask to bitstreams from our other encoders (ASCII or Embedding) reduces output-side recoverability. Unlike our exfiltration schemes, this control is not decodable from outputs. It removes decodable information from the text and isolates whether a detector can identify steganographic behavior from internal signatures *alone*.

Table 3. XOR Encoding Detection: A-XOR denotes ASCII encoding with XOR, while E-XOR denotes embedding encoding with XOR. Bit=bit accuracy (%), Cos=cosine similarity. Green = lower 95% CI above upper 95% CI of base (detection possible).

		Llama-8B			Ministral-8B		
Enc	M	Base	LoRA	Full	Base	LoRA	Full
A-XOR+E	Bit	67±4	99	97±1	64±5	99	72±4
	Cos	.35±.00	.46±.00	.52±.01	.26±.00	.50±.00	.24±.01
E-XOR+E	Bit	61±3	63±3	64±4	60±4	66±5	59±3
	Cos	.34±.00	.42±.00	.49±.00	.25±.00	.51±.00	.22±.01

Table 3 demonstrates that even in this extreme case, where we cannot recover the code derived from the secret, exfiltration still leaves a detectable mechanistic trace. Specifically, we observe that, outside of Ministral-8B full fine-tune, cosine similarity for embedding recovery is consistently higher in fine-tuned models. Crucially, this occurs despite testing on OOD prompts. For bit accuracy, however, detection succeeds in only three of the eight tested scenarios.

8. Conclusion

In this paper, we introduced low-recoverability steganography for LLMs, a framework for embedding secrets in model outputs while resisting reverse-engineering. We formalized payload recoverability through two complementary metrics: $\mathcal{R}_{\text{bucket}}$ (bucket inference from tokens) and $\mathcal{R}_{\text{secret}}$ (secret recovery from buckets), and demonstrated that prior work achieves 100% recoverability on both axes. Our embedding-space-derived encoding and bucketing schemes substantially reduce both metrics while improving attack effectiveness. On Llama-8B (LoRA), Ministral-8B (LoRA), and Llama-70B (LoRA) we observed exact secret recovery improvements of up to 17→30% (+78%), 24→43% (+80%), and 9→19% (+123%), respectively. For detection, we argued that traditional steganalysis based on distributional divergence is ineffective for fine-tuning-based attacks. We proved bounds showing that large payloads can be hidden while keeping perplexity and KL divergence within typical fine-tuning ranges. Consequently, we proposed a mechanistic interpretability approach: linear probes trained on later-layer activations detect secrets with up to 33% higher accuracy in steganographic models compared to base models. Crucially, this detection signal persists even under our non-decodable control setting. Our detection method assumes access to model hidden states; extending these findings to output-only or black-box settings remains an important direction for

future work. Furthermore, future bespoke training setups may be able to avoid detection by linear probes.

Impact Statement

This paper presents work on steganography in large language models, which has potential dual-use implications. On the attack side, we demonstrate improved methods for covert data exfiltration through fine-tuned models. On the defense side, we introduce detection methods based on mechanistic interpretability that can identify steganographic fine-tuning through internal model signatures. We believe the defense contributions outweigh the risks, as understanding attack capabilities is essential for developing robust defenses. Our detection methods provide a foundation for identifying malicious fine-tuning in deployed models. We will release the full detection code; on the attack side, we will coordinate with model providers for responsible disclosure.

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